

Week 11: Routing with React Router

1. Setting up React Router v6

React Router is a powerful library used in React applications for navigation between different pages without refreshing the browser. During this week, React Router v6 was set up and configured to manage page transitions efficiently. It allows developers to build single-page applications with multiple views while maintaining smooth navigation. Proper setup of routing is important for improving application structure and user experience.

2. Defining Routes, Link, NavLink

Routes define which component should render for a specific URL path. The `Link` component is used for navigation between pages without reloading the browser. `NavLink` works similarly to `Link` but provides active styling for currently selected routes. These features make navigation easier and improve usability in larger React applications.

3. Route Parameters, Nested Routes

Route parameters allow passing dynamic values in URLs, such as user IDs or product IDs. This helps create dynamic pages based on different data. Nested routes are used when multiple related pages exist under a common parent route. This structure improves organization and makes routing more scalable in complex applications.

4. Protected Routes

Protected routes restrict access to specific pages based on authentication or authorization. For example, a dashboard page may only be accessible to logged-in users. This ensures better security and user management in web applications. Implementing protected routes is essential for real-world React projects.

Conclusion

This week focused on routing in React applications using React Router v6. By learning route setup, navigation components, route parameters, nested routing, and protected routes, application flow and page management became more efficient.